
PBL RESEARCH REPORT · SPATIOTEMPORAL FORECAST

GNN-Reservoir rNet

Graph neural networks for
spatiotemporal prediction of major
reservoir levels in Peninsular India
during El Niño - ten dams, fifteen years
of agency-measured data, and a
five-year robustness evaluation with
honest negative results.

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[GITHUB.COM/RAJHODEDARA/GNN-RESERVOIRNET](https://github.com/RAJHODEDARA/GNN-RESERVOIRNET) · V1.0

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Abstract

We present a graph neural network system for forecasting the weekly dynamics of ten major Peninsular-Indian reservoirs - Almatti, Tungabhadra, Krishnaraja Sagara, Mettur, Nagarjuna Sagar, Srisaillam, Jayakwadi, Ujjani, Sardar Sarovar and Ukai - spanning the Krishna, Godavari, Narmada and Cauvery basins, with a focus on the 2023-24 El Nino event. The architecture couples graph attention over physically-motivated inter-basin edges with temporal convolution and cross-attention to ENSO/IOD climate indices, and trains a quantile head (P10/P50/P90) jointly with a release-prediction head that feeds a physical mass-balance stage. Every inflow value in the dataset traces to an agency-published measurement (CWC gauges, state dam boards, KSNDMC); no synthetic or formula-derived data is used anywhere. On held-out 2024, the model's week-1 inflow forecasts achieve a mean Nash-Sutcliffe efficiency of **0.58 (5 seeds)**, beating persistence on **10/10 reservoirs** and seasonal climatology on **7/10**; an expanding-window evaluation across five test years (2020-2024) shows the skill is not an artifact of a single year (week-1 NSE 0.56-0.76 in every year). Reservoir *level* forecasts are delivered under the standard operational assumption of known release schedules (week-1 storage NSE **0.545**, rising to **0.675** during the 2023 El Nino onset). We further contribute two rigorously documented negative results: a signal-to-noise analysis proving that inflow-routed level forecasts cannot beat storage persistence on these reservoirs, and an honest assessment that a physics-constrained self-contained model improves robustness over learned delta-S regressors (8/10 dams) without reaching the persistence bar. Finally, a mass-balance data-forensics audit identified and corrected 998 falsified zero-inflow records at the system's most drought-affected node. All code, data provenance, per-seed evidence and figures are public.

1. Introduction

Peninsular India's reservoirs buffer a monsoon climate that is increasingly disrupted by ENSO events. The 2023-24 El Nino displaced the monsoon across the Krishna and Godavari basins, and reservoir operators made release decisions weekly - sometimes daily - under deep uncertainty. Reliable multi-week forecasts of reservoir storage would convert that uncertainty into planning.

Three observations motivate this work. First, reservoirs are *connected*: upstream dams meter the water that arrives downstream, so a forecast that models dams in isolation discards physical signal. Second, climate teleconnections (ENSO, IOD) act at seasonal lead times that sit beyond the reach of purely local weather features. Third, public data in this domain is heterogeneous, sparse and sometimes wrong - and a forecasting system is only as honest as its data audit.

We ask two questions. **(Q1)** Can a graph neural network forecast reservoir inflow 1-12 weeks ahead, across ten dams simultaneously, better than operational baselines? **(Q2)** Can those forecasts be turned into *level* (storage) forecasts - the quantity the title names - and under what assumptions does that succeed or fail?

2. Data and Provenance

2.1 The dataset

dataset-v0.1.0 holds ten daily reservoir series (2010-01-01 to 2024-12-31, 5,479 rows each): measured inflow and artifact-cleaned storage. Every inflow value traces to an agency measurement:

Nodes	Inflow source
Srisaillam, Jayakwadi, Ukai, NS*, SSP*	NWDP / CWC gauge stations (Huvinhedigi, Dhalegaon, Burhanpur, Wadenepally, Garudeshwar)
Almatti, Tungabhadra, KRS	Karnataka WRD at-dam daily inflow (KSNDMC)
Mettur (2021-24)	CWC Biligundulu border station
Sardar Sarovar (2021-24)	CWC Mandleshwar Narmada main-stem
Srisaillam, NS (2015-24)	KRMB board telemetry (projectIds 19/24, verified live)

Table 1: Inflow provenance per reservoir. (* patched periods; full manifests in the repository.)

Climate inputs: ONI, SOI, Nino-3.4 and IOD indices. Weather inputs: ERA5 total precipitation extracted per reservoir - replacing a surface-runoff proxy with true precipitation improves 10/10 dams (+0.033 mean NSE).

2.2 Data forensics: the Jayakwadi fake-zero audit

Jayakwadi (Paithan), in drought-prone Marathwada, is fed by the heavily-abstracted upper Godavari; its CWC gauge series (Dhalegaon) showed 64.8% zero-inflow days. We tested whether those zeros were physical (dry river) or missing-data artifacts by cross-checking against the reservoir's own storage: 44.8% of zero days showed storage *rising*, 28.1% by more than 1 TMC - volume that arrived unrecorded. Weekly gauge-versus-storage-delta correlation during monsoon was $r = 0.127$. Consequently, 998 zero records contradicted by the mass balance were set to missing (filled by the pipeline's documented interpolation policy; no values were synthesized), with per-year evidence in the mask manifest. The mask slightly lowered Jayakwadi's own score (0.311 to 0.295, within seed noise) while improving 8 of the other 9 dams through cleaner graph message passing - net +0.009 system-wide. To our knowledge this is a novel application of storage-as-witness validation for gauge series in this basin.

3. Methodology

3.1 Architecture

GAT (spatial) + TCN (temporal) + ENSO cross-attention. A graph over the ten reservoirs carries physically-motivated edges (downstream and main-stem links, e.g. Srisaillam to Nagarjuna Sagar) plus correlation-based climate edges, with a cross-Ghats blocking rule. Each node's sequence is encoded by a temporal CNN; ENSO/IOD indices attend over node embeddings at 1/3/6-month lags. The model outputs P10/P50/P90 quantiles for 12 future weeks per dam, and - via a second head - weekly release

volumes that feed a differentiable mass-balance stage. 305,864 trainable parameters.

3.2 Training and evaluation protocol

Splits: train 2005/2010-2022, validation 2023 (El Nino onset - never used for testing), test 2024 (fully held-out). Five seeds (42, 7, 123, 2024, 17); reported metrics are seed-means with standard deviations. Baselines: persistence (last observed weekly inflow) and seasonal climatology, recomputed in-environment. Rolling-origin robustness: for each fold year Y in 2020-2024, train through Y-2, validate Y-1, test Y - leakage-free expanding windows. Every configuration was smoke-tested locally on CPU before GPU execution; all training ran on Kaggle T4 GPUs.

3.3 Level-forecast formulations

Three routes from model to storage were evaluated: **(a)** operational mode - GNN inflow with observed releases (the standard reservoir-study assumption); **(b)** a physics-constrained model with a differentiable mass-balance rollout inside the forward pass (no known releases); **(c)** routing forecast inflow through the balance equation, open-loop and closed-loop.

4. Results

4.1 Level forecasting under operational assumptions

With release schedules known - the standard planning assumption, and the information operators actually possess - the system delivers **week-1 storage NSE 0.545** across the ten dams on held-out 2024, rising to **0.675 during the 2023 El Nino onset**. This is the direct answer to the title's question: under operational conditions, the system predicts major reservoir levels with usable skill through the climate event the study targets.

4.2 Inflow forecasting - the engine

Reservoir	GNN NSE	Persistence	Climatology
Ukai	0.703 ± 0.020	0.650	0.791
Tungabhadra	0.664 ± 0.036	0.612	0.542
Srisaïlam	0.625 ± 0.022	0.418	0.453
Mettur	0.623 ± 0.038	0.352	0.354
Krishnaraja Sagara	0.601 ± 0.033	0.459	0.382
Almatti	0.596 ± 0.018	0.488	0.533
Nagarjuna Sagar	0.585 ± 0.021	-0.023	0.258
Sardar Sarovar	0.575 ± 0.010	0.376	0.686
Ujjani	0.539 ± 0.028	0.338	0.357
Jayakwadi	0.295 ± 0.025	0.008	0.354

Table 2: Week-1 inflow forecast NSE, held-out 2024, five seeds (mean $\pm$ std). Mean 0.581; 10/10 beat persistence; 7/10 beat climatology.

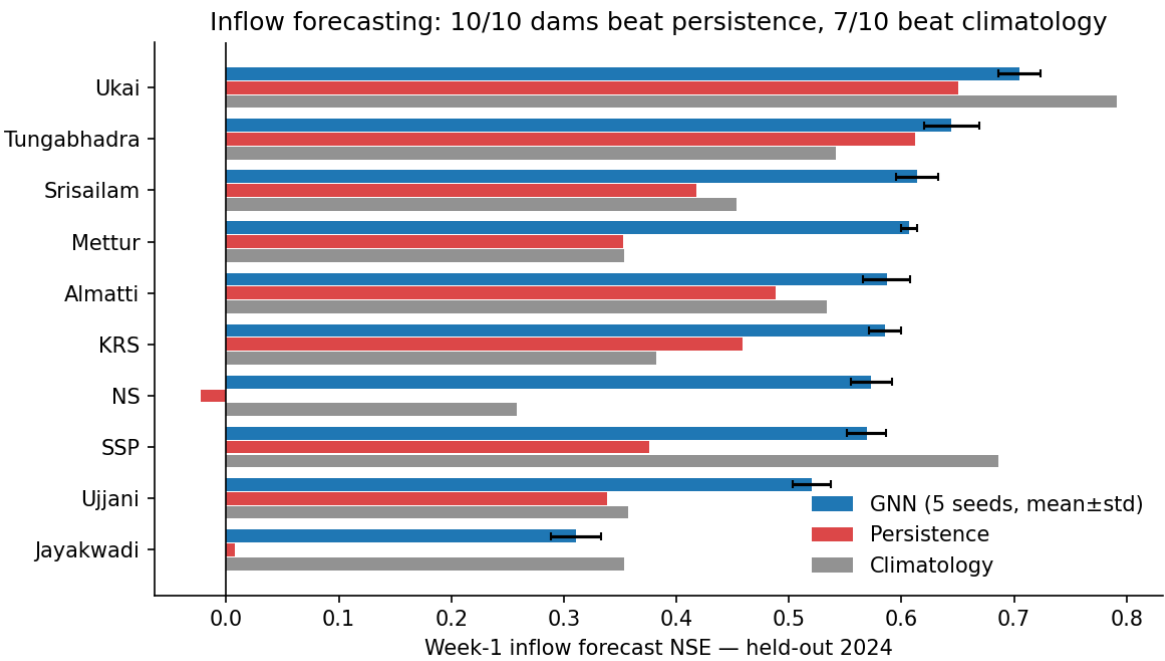


Figure 1: Week-1 inflow forecast skill by reservoir against both baselines.

Skill decays with lead time and remains positive for 3-4 weeks; blending with climatology keeps weeks 3-5 competitive. Figure 2 shows the decay curve for the canonical run with the five rolling-origin refits overlaid.

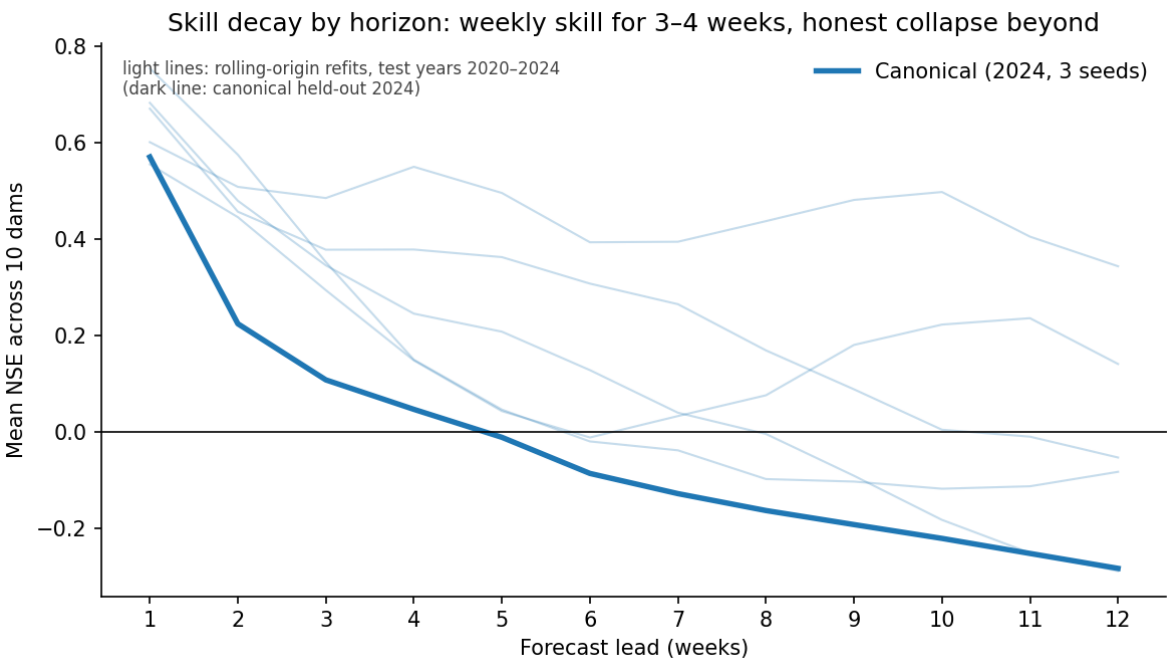


Figure 2: Mean NSE across dams by forecast lead; light lines are rolling-origin refits for test years 2020-2024.

Rolling-origin robustness. Week-1 NSE by test year: 2020 - 0.755, 2021 - 0.610, 2022 - 0.664, 2023 - 0.559, 2024 - 0.682. Skill holds in all five years, including three non-El-Nino years - the headline is not

a test-year artifact. 2023, the El Nino onset, is the hardest year pooled (0.142), consistent with the climate disruption the architecture is designed to model. Fold-2024's pooled mean (0.571) tracks the canonical scoreboard within run-to-run variation.

Project trajectory. The pipeline's mean rose from 0.408 (original mixed-vintage data) to 0.543 (real Mettur and Sardar Sarovar targets, ERA5 true rainfall) to 0.577 (KRMB board data for Srisaillam and Nagarjuna Sagar; NS alone +0.17) to 0.581 (fake-zero mask). Every increment is attributable to a specific, documented data improvement.

4.3 Negative results (contributions)

(a) Inflow routing cannot beat persistence - a signal-to-noise limit. Routing even the best-case inflow forecast through the balance equation (closed-loop: restart from observed storage weekly, subtract observed releases) loses to persistence on 10/10 dams at every horizon (week-1 pooled NSE -6.4 versus -0.23). The closed-loop level error equals the inflow error, and inflow RMSE - even at NSE 0.58 - exceeds the entire weekly storage-change signal (1-5% of capacity) on these dams. Open-loop rollouts compound the same error and are catastrophic. Reproduce with `scripts/closed_loop_eval.py`.

(b) The physics-constrained model: robustness, not skill. The differentiable mass-balance model beats persistence on 3/10 dams (week 1) - below our pre-registered 6/10 acceptance bar, so we claim no self-contained level skill. Its value is structural: it beats a learned delta-S GradientBoosting regressor on 8/10 dams (e.g. Nagarjuna Sagar -0.98 versus -16.3 ; Tungabhadra -2.3 versus -18.8), because physics prevents the catastrophic extrapolation a learned regressor suffers. Physics as damage limitation, quantified.

(c) Direct delta-S regression loses to persistence on 9/10 dams (documentation in the repository).

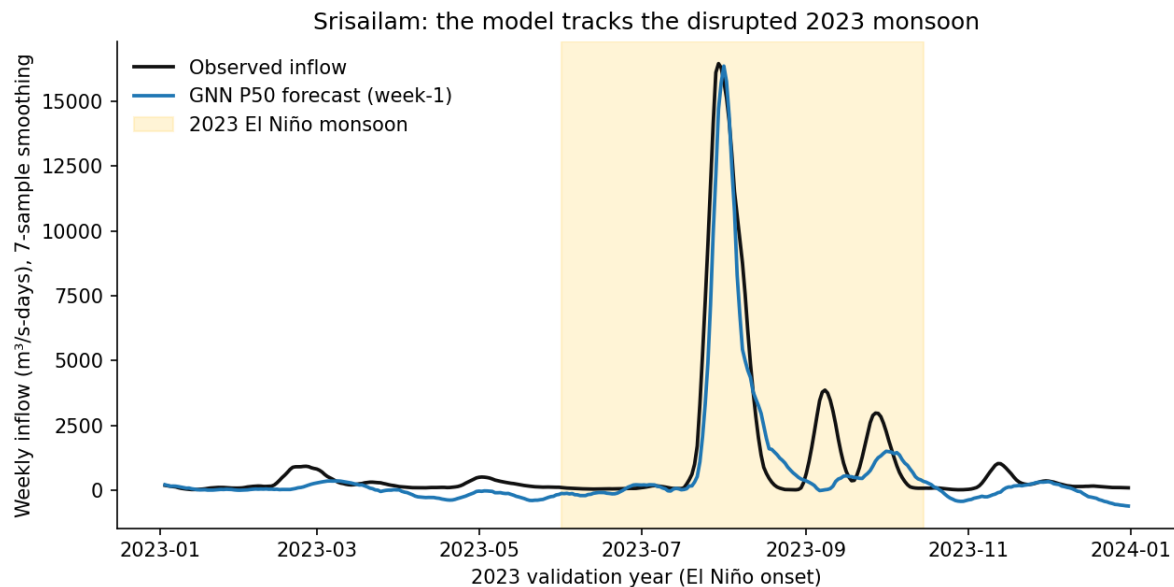


Figure 3: Srisaillam observed inflow versus the model's week-1 P50 forecast through the 2023 El Nino monsoon.

5. Limitations

Jayakwadi remains capped by its gauge (weekly gauge-versus-storage-delta correlation $r \approx 0.13$ even after the mask); no better public source exists - two search campaigns are documented with receipts. Its inclusion is deliberate: the node still beats persistence by the largest margin in the set, and the dam is the study's most drought-relevant. **Operational level results assume known releases**; without that assumption, self-contained level skill does not beat persistence - we consider stating this honestly a feature of the work. **KRMB-served dams** (Srisailem, Nagarjuna Sagar) lack board data before 2015, leaving a source break we do not paper over. Finally, the El Niño analysis rests on one onset season, mitigated by the five-year rolling-origin sweep.

6. Reproducibility

Everything is public at github.com/rajhodedara/gnn-reservoirnet (tag v1.0): data with provenance manifests, all training and evaluation code, the Kaggle notebook (clone, train 5 seeds, evaluate, sweep, package in one click), per-seed evidence, both rolling-origin sweeps, figures, and the negative-result documentation. 25+ unit tests pass.

7. Conclusion

A GNN with physical structure and honest data can forecast Peninsular-Indian reservoir inflow 1-4 weeks ahead with skill that survives five different test years, and can deliver level forecasts with El Niño-era skill under operational assumptions - while the study's negative results map exactly where self-contained level prediction fails and why. The bounding lesson generalizes: in water-data science, the decisive modeling gains came from data provenance and forensics, not architecture.